

UNIT-3: PROJECT PLANNING

What is the Initiation phase in project management?

In project management, the key to success is a calculated beginning, and this is where project initiation excels. It serves as the compass for your project, making sure that every move you take and every objective you set is reachable. But what precisely is project initiation, and why is it important for the SEO success of your project?

In this article, we're going to break down project initiation in simple terms. We'll explore why it's so important, what it involves, and how to do it right. Whether you're a complete beginner or just looking to brush up on your project management skills, follow along as we uncover the secrets to starting your projects off on the right foot.

Project initiation is like the starting point of a project journey. It's when you decide what you want to do, why you want to do it, and who will be involved. It's all about getting everyone on the same page before you dive into the details of how to make it happen. The initial stage in beginning a new project is called project initiation. Establishing the project's purpose and the commercial value it will provide helps you gain support from important stakeholders.

Benefits of a Well-Executed Initiation Phase:

1. **Clear Project Direction:** During the beginning stage, it's important to set clear goals for a project. This helps guide the team and keeps them focused in one direction.
2. **Risk Mitigation:** By checking things carefully, we can find the possible dangers and problems right at the start of a project.
3. **Resource Optimization:** Starting things right means better use of stuff. By knowing the project aims, managers can figure out what talents and tools are needed.
4. **Stakeholder Alignment:** The starting stage helps to find and involve top people. This is an important part for all involved in something. Getting everyone on the same page and knowing what is expected during this part of a project helps people support it. This happens if you talk clearly with them about these needs from start to finish in their projects.

Uses of the Initiation Phase:

1. Feasibility Analysis

The start part needs a good look at if the project can work. You check stuff like money, computer tech and day-to-day operations Issues that make it happen or not. This study helps to see if the project works and follows what the company is trying to achieve in their big plans.

2. Project Charter Development

From the start phase, a major result is the project charter. This paper tells what the project is about, its goals and who's involved. It also shows the first needs for resources needed to start it up nicely like a guidebook during all the actions of ideal efforts.

3. Goal Definition

Setting up goals for a project is a key job in the starting phase. This means making clear the aim of a project, and setting specific goals that can be measured and accomplished. They should line up with when they need to get done. This will lead on through what you make doing inside it runs its life-cycle work within teams too.

4. Initial Planning

In the beginning, you make an early project plan. Later on, more precise planning is done to go forward with it fully and properly. However, at first, this general idea for a new thing is thought out quietly alone before anyone else does any work or decides whether something can happen easily without problems happening in the different steps each time: what will be accomplished? Who'll do the tasks? When they.

Project Initiation Vs Project Planning

Here's a comparison of Project Initiation and Project Planning in tabular form:

Aspect	Project Initiation	Project Planning
Objective	Define the project's purpose, goals, and scope	Develop a detailed roadmap for achieving goals
Focus	Why the project is needed and what it aims to achieve	How the project will be executed and managed
Activities	Identifying stakeholders, defining objectives, and establishing a project charter	Breaking down tasks, setting timelines, and allocating resources
Key Outputs	Project charter, initial budget, and high-level timeline	Detailed project plan, task lists, and resource allocations
Importance	Sets the foundation and direction for the project	Guides the execution and management of the project

What is Project Scope Management Plan?

A plan for Managing Projects is very important in handling the projects. It helps set goals and makes sure things are done the right way during project work. This plan talks about important things like goals, people involved, and what needs to be done. It also includes a timetable, a money budget, and ways of handling risks. This makes sure changes are kept under control so the project does well. A basic part of managing what a project does, it helps to properly run and look after projects.

Purpose of a Scope Management Plan



1. Clarification of Project Goals

- The Plan for managing the scope sets up rules on, how to clearly show and also explain what goals the project wants.
- It ensures that the team members on a project know and are in agreement about what they must do.

2. Work Definition and Boundaries

- The plan sets standards for making, testing and watching the amount of work in a project.
- This helps to define the boundaries for a task and halt it from turning too large or exceeding its targets.

3. Risk Mitigation

- By making clear rules, the plan helps in lessening dangers caused by changing what's included or the uncertain goals of a project.
- It gives a clear way to find, check, and look after the risks related to the project scope.

4. Optimal Resource Utilization

- The plan for managing scope makes sure, that we use project stuff properly.
- It makes sure resources are given rightly, so project goals can be met by setting up the boundaries.

5. Client and Stakeholder Alignment

- It helps customers, those involved and the teams who work on projects agree with each other because it explains what the project is for.
- When people love something, they are told about it and everyone gets involved. This stops any misunderstandings or disagreements.

6. Adaptability and Monitoring

- The plan is an essential thing that can be altered as the circumstances of a project change.
- This is useful for monitoring the project updates and highlights the key points, which allows for speedy adjustments.

7. Foundation for Effective Management

- The Scope Management Plan is a main document that helps in managing the project things properly.
- It offers a clear instruction for project leaders to manage and watch the size of their task during its various steps.

Work Breakdown Structure (WBS)

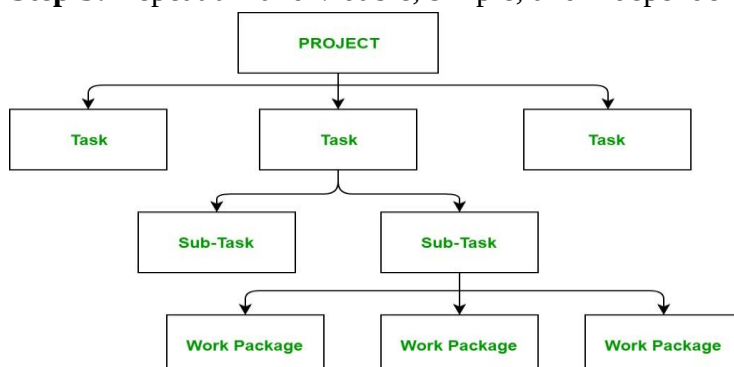
A Work Breakdown Structure includes dividing a large and complex project into simpler, manageable, and independent tasks. The root of this tree (structure) is labelled by the Project name itself. For constructing a work breakdown structure, each node is recursively decomposed into smaller sub-activities, until at the leaf level, the activities become undividable and independent. It follows a Top-Down approach.

Steps Work Breakdown Structure:

Step 1: Identify the major activities of the project.

Step 2: Identify the sub-activities of the major activities.

Step 3: Repeat till undividable, simple, and independent activities are created.



Construction of Work Breakdown Structure:

1. Firstly, the project managers and top level management identifies the main deliverables of the project.
2. After this important step, these main deliverables are broke down into smaller higher-level tasks and this complete process is done recursively to produce much smaller independent tasks.
3. It depends on the project manager and team that up to which level of detail they want to break down their project.
4. Generally, the lowest level tasks are the most simplest and independent tasks and takes less than two weeks' worth of work.
5. Hence, there is no rule for up to which level we may build the work breakdown structure of the project as it totally depends upon the type of project we are working on and the management of the company.
6. The efficiency and success of the whole project majorly depends on the quality of the Work Breakdown Structure of the project and hence, it implies its importance.

Uses of Work Breakdown Structure

1. **Cost estimation:** It allows doing a precise cost estimation of each activity.
2. **Time estimation:** It allows estimating the time that each activity will take more precisely.
3. **Easy project management:** It allows easy management of the project.
4. **Helps in project organization:** It helps in proper organization of the project by the top management.

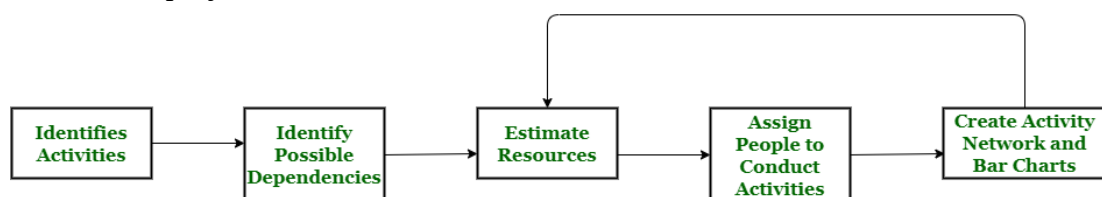
What is Effort Estimation in Software Development?

Effort estimation is a process that forms part of the software development life cycle and is key to the assessment of the probable number of hours that may be required for the accomplishment of particular tasks in a software development project. In high-level definition, effort estimation in software development is the process of quantifying the amount of work that can be done in terms of person-hours or person-days needed to accomplish a given task or the whole project. This involves, i.e., the whole process of the SDLC, that is, the gathering of requirements and preparation of specifications, the design, the coding and testing of the software, and the maintenance of the software. Effort estimation is an essential part of project management, as it enables one to predict the amount of time necessary to complete a project and coordinate the costs and quality of a project to meet customer expectations.

Effort estimation is a fundamental task in software development, which is crucial in managing and controlling the productivity process within a project as well as within the software production organization in general.

Project Scheduling

A schedule in your project's time table actually consists of sequenced activities and milestones that are needed to be delivered under a given period of time. **Project schedule** simply means a mechanism that is used to communicate and know about that tasks are needed and has to be done or performed and which organizational resources will be given or allocated to these tasks and in what time duration or time frame work is needed to be performed. Effective project scheduling leads to success of project, reduced cost, and increased customer satisfaction. Scheduling in project management means to list out activities, deliverables, and milestones within a project that are delivered. It contains more notes than your average weekly planner notes. The most common and important form of project schedule is Gantt chart.



Project Scheduling Process

Process: The manager needs to estimate time and resources of project while scheduling project. All activities in project must be arranged in a coherent sequence that means activities should be arranged in a logical and well-organized manner for easy to understand. Initial estimates of project can be made optimistically which means estimates can be made when all favorable things will happen and no threats or problems take place. The total work is separated or divided into various small activities or tasks during the project schedule. Then, Project manager will decide time required for each activity or task to get completed. Even some activities are conducted and performed in parallel for efficient performance. The project manager should be aware of fact that each stage of project is not problem-free.

Problems arise during Project Development Stage:

- People may leave or remain absent during particular stage of development.
- Hardware may get failed while performing.
- Software resource that is required may not be available at present, etc.

The project schedule is represented as set of chart in which work-breakdown structure and dependencies within various activities are represented. To accomplish and complete project within a given schedule, required resources must be available when they are needed. Therefore, resource estimation should be done before starting development.

Resources required for Development of Project:

- Human effort
- Sufficient disk space on server
- Specialized hardware
- Software technology
- Travel allowance required by project staff, etc.

Advantages of Project Scheduling:

There are several advantages provided by project schedule in our project management:

- It simply ensures that everyone remains on same page as far as tasks get completed, dependencies, and deadlines.
- It helps in identifying issues early and concerns such as lack or unavailability of resources.
- It also helps to identify relationships and to monitor process.
- It provides effective budget management and risk mitigation.

Estimating Activity Resources:

Effort estimation approaches can be broadly categorized into the following:

- **Expert Judgment:** This approach is based on the perception of experts or some particular teams and groups to make an estimation of the effort that will be required. It is commonly used for the purpose of getting and consolidating the views of experts, including the use of Delphi and Wideband Delphi.
- **Analogous Estimation:** This method is similar to the preceding one in that historical data is also analyzed to establish the effort of a current project. It presumes that the present project will go through formative processes just like past projects that were implemented.
- **Parametric Models:** These models employ mathematical equations through which effort is forecasted with the aid of project characteristics such as size, level of difficulty, and the expertise of the staff to be enlisted. Traditional forecasting models include COCOMO, which falls into the category of parametric models.
- **Function Point Analysis (FPA):** FPA has been defined to reflect the extent of functionality that is delivered to the user, and this is the measure that FPA uses to derive effort. Effort can be too complicated, where function points are counted, including inputs, outputs, and/or queries, and conversion rates are used.
- **Story Points and Agile Estimation:** Also in Agile methodologies, activity valuation is done in terms of story points, which is a measure of effort to implement a particular user story. This ensures that team members attain consensus when estimating the story points using various techniques, such as planning poker.

- **Machine Learning-Based Estimation:** Automated methods apply historical data from completed projects to estimate effort using machine learning models. These models can also be refined and modified as more 'data', that is, more measuring, is carried out.

Selection of Estimation Approaches

Choosing the right estimation approach depends on various factors, including:

- **Project Size and Complexity:** Larger, more complex projects may benefit from parametric models or machine learning-based approaches, while smaller projects might be effectively estimated using expert judgment or analogous estimation.
- **Data Availability:** If historical data is available, analogous estimation or machine learning models can provide more accurate estimates. In the absence of such data, expert judgment may be more reliable.
- **Development Methodology:** Agile projects often use story points and techniques like planning poker, while traditional projects may rely more on parametric models and function point analysis.
- **Team Experience:** The expertise and experience of the team can significantly influence the accuracy of expert judgment-based estimates.

Assessing the Accuracy of Estimates

Assessing the accuracy of effort estimates involves comparing estimated effort with actual effort after project completion. Common techniques include:

- **Variance Analysis:** Calculating the difference between estimated and actual effort to identify deviations and understand their causes.
- **Statistical Measures:** Using metrics such as mean absolute error (MAE), mean squared error (MSE), and mean magnitude of relative error (MMRE) to quantitatively assess estimation accuracy.
- **Post-Mortem Analysis:** Conducting thorough reviews at the end of projects to evaluate estimation performance and identify areas for improvement.

Estimating Activity Durations:

Before you begin developing the schedule of a project, you need to know the expected durations of all activities of a project. Project management frameworks such as the PMI's PMBOK contain different tools and techniques for estimating durations. The expected amount of time for the completion of an activity is the basis for developing the project schedule. It can also serve as an input for estimation of the cost of an activity, depending on the type of the activity and the estimation technique.

What Are Activity Duration Estimates?

The purpose of estimating activity durations is to determine the amount of time it takes to complete an activity. Estimate activity durations is a process of the Project Schedule Management knowledge area according to PMI's Guide to the Project Management Body of Knowledge (PMBOK®, 6th ed., ch. 6.4).

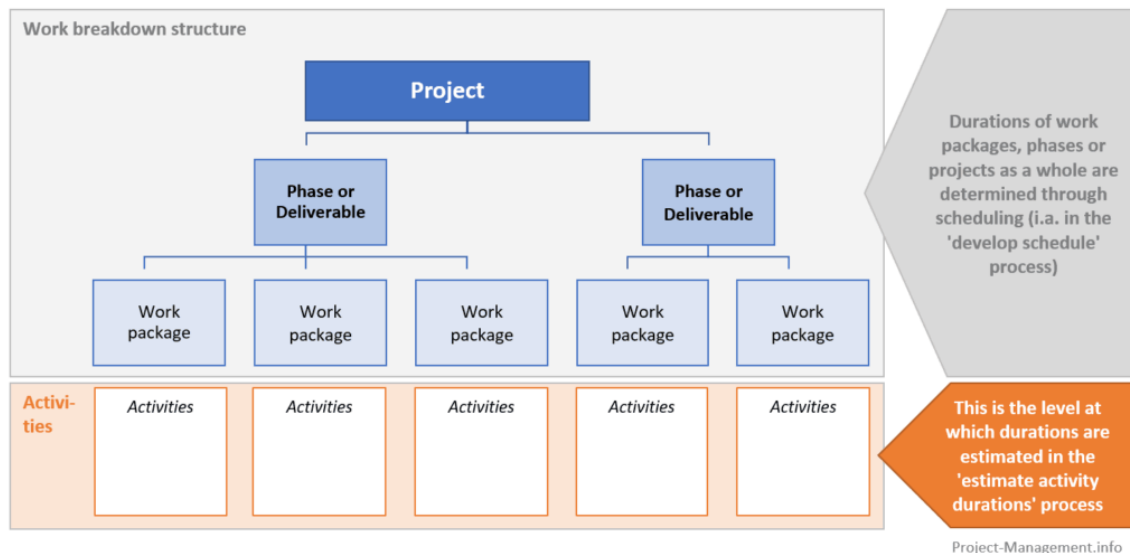
This process requires several input parameters, which include, in particular, the scope of work and the list and characteristics of planned activities as well as the resources that are deployed to perform the work.

The PMBOK lists

- expert judgment,
- analogous estimating,
- parametric estimating,
- bottom-up estimating, and
- three-point estimation

as techniques to determine duration estimates. Read on for an overview of these estimation techniques.

The estimation of durations is normally done on the level of activities (see below image). Determining the total duration of a work package or the whole project requires scheduling of activities, taking their individual durations into account.



This includes the consideration of dependencies between activities and often involves the development of a critical path (the longest chain of activities to achieve the desired outcome). It is therefore typically not accurate to simply calculate the sum of activity durations if you determine the duration of work packages or whole projects. This is because the sum of activity durations may overstate (sometimes understate) the total time to completion of a project as some activities can run in parallel while others are dependent on the completion of a preceding activity.

What Is the Basis of (Duration) Estimates?

The basis of estimates is a documented set of assumptions and constraints that underlay a duration estimate. This document also describes the methods applied to produce an estimate, the range of potential outcomes as well as the confidence of the estimate(s).

This documentation is important in projects for several reasons. It makes the constraints and assumptions transparent that have been considered for the estimation. Once an assumption becomes invalid, this may trigger the need for a new estimation. A project manager can also use these documents to ensure consistency across the different activities in a project, as a justification for the project schedule and for the communication with stakeholders.

How Is Activity Duration Estimated?

The estimation process requires some input information and documents and implies the use of one or multiple estimation techniques. The following subsections elaborate on these requirements.

The estimation of activity durations should generally be done by the person who is the most familiar with the activity's type of work. This is even more important if an estimation technique is used that relies on the expertise and experience of the estimator(s) instead of historical data.

The output of duration estimations is always a number of time periods (e.g. days, weeks, months; source). The level of confidence and whether it is a single estimate or a range depend on the estimation technique that is applied.

Planning and scheduling with a Gantt chart

Planning in project management is a necessity. And so are Gantt charts. So, whether you are new to project planning or already an experienced project manager, it's likely that you've heard of these tools. In fact, it's also likely that you've already done some project scheduling with a Gantt chart.

Most of today's standalone web-based software has the advantage of having critical project management features, including the Gantt chart, readily available. Which is great in today's challenging and dynamic work environment. All you need to do is choose the right tools.

Just imagine being a project manager hundreds of years ago, and trying to build the Taj Mahal or Sagrada Familia without modern planning tools. No wonder they took hundreds of years to complete!

Nowadays, it's possible to use modern Gantt chart planning to better manage your project resources and control for unexpected situations. There's a reason why we still use this solution. Because even after a century, we still haven't been able to think of a better alternative.

There are several other Gantt chart features that can make scheduling easier and quicker:

- **Color schemes** – use it for tasks (combine tasks by colors), for projects (use different colors for different projects) or for stages of projects (red for “canceled,” green for “completed”). An

overview of your project is one of the keys to success, and color coding your project management gives you a quick assessment of your needs and metrics. It'll bring you one step closer to project success.

- **Custom Data Fields** – customize your projects, tasks, people, and other resources by their metrics that are important for their projects. It could be their skills, the year when certain machinery was bought, or the name of a project manager. This allows for your tasks and bar graphs to handle way more information. Your graphs are more detailed and you can then use this info in your planning and reporting.
- **Notes and history** – you can write notes for the tasks and see the history of changes in tasks. This helps you avoid mistakes by keeping track of who made changes and to what. It is like a sticky note on your whiteboard... but happening in real-time, so everyone can see it!
- **Utilization** – planning allows you to use your time and resources more efficiently. In Gantt you have an overview of the utilization of ALL your people and resources. This lets you ensure that nothing is overlooked or gives you the option of taking unused capacity from another project. Efficient planning is growth planning.
- **Reports** – sometimes not only the graph but the tables of calculation can be helpful, so you can actually calculate the hours, utilization, durations and many more. Plus, a lot of times you can choose to even automate your reports, so you'll get them at the time interval you want. Let the program do the work for you, so you can focus on other things.
- **Flexible timeline** – since Gantt charts are not drawn manually on paper anymore, there are programs that let you play around in your timeline. If you need to see today's plans, just zoom in. When you need to understand the next month's plans or schedule next year's project, just zoom out. You are the project manager who handles your projects, not the other way around!

What is Gantt Chart in Project Management?

There are many definitions of a Gantt chart. But at its core we can say that they are a tool used to illustrate project progress, deadlines and elements.

- A Gantt chart is a visual project management tool that helps to plan and schedule projects of every size.
- Gantt charts look like a horizontal bar chart that shows project management timelines, task starting and ending dates, dependencies between different tasks, and general project task flow.
- It is a visual interpretation of the project which gives an overview of the project's progress, timeline, and tasks over its entire time frame.

The Projects usually contain multiple tasks and activities. As a Project Professional, when you want to manage a complex project, it can be not easy to track the tasks, their relationships, duration, and status. A single change in one task can affect the progression of a project and thus, it is important to understand the inter-relationship among the different tasks. Gantt Chart helps us manage the timelines of the project. This article covers a detailed guide on the Gantt Chart, its features, and use cases so that you can manage your project execution efficiently.

What is the Gantt Chart?

A Gantt Chart is a visual project management timeline that consists of horizontal bars, usually color-coded.

1. These bars show all the project tasks, when they are due, who is responsible, and how long each task will take.
2. Therefore, Gantt charts illustrate the start and finish dates of the terminal elements and summary elements of a project.
3. In other words, the Gantt chart assembles all pieces of tasks that are relative to a fixed deadline

Basic components of a Gantt chart

Gantt charts could initially appear difficult. However, if you can interpret them, you'll be able to determine precisely where your initiatives stand and what has to be done to steer them towards success. Understanding how the various components work together to create a Gantt chart is the key to reading one. **Here are the following components of a Gantt Chart:**

- **Projects dates and timeline** – this gives project managers an overview of all the project dates. From the start, to dates connected with project tasks, and to the finish. You have the complete time frame at your fingertips!
- **Gantt chart bars as project tasks** – projects normally consist of different tasks and the Gantt chart is a great way to see them all in one place. A visual overview helps you make sure that everything is in the right place on the timeline and nothing is forgotten. Task names are normally set on the Gantt chart. Luckily, modern charts are made so that you can easily change the location and the length of the task bar.
- **Milestones** – milestones are the little “wins” of the projects. They are normally at the end of the task and hold some significance for the project. Usually, milestones are displayed as diamond-shaped – which feels like a bonus after the completion of the task. *ProTip:* Always celebrate your little “wins”. They help to break your project into more achievable pieces.
- **Dependencies** – there are always tasks in your projects that need to be completed before the next task can begin or end. So one task is dependent on the other’s start or finish. In Gantt charts, dependencies between the tasks are shown with little arrows. *ProTip:* dependencies can help you focus on the base tasks that are directly connected with the general project flow.
- **Resources** – In a classic Gantt chart, you can add a resource to your task to see who’s responsible for the task and who’s working on it. In modern Gantt charts, there are ways to make it more visual and understandable. It’s not rare to use Gantt charts for resource planning and project portfolio management, in addition to single project planning.

Adding Milestones to Gantt Charts:

Adding milestones to Gantt charts highlights key dates, deliverables, or phase completions, generally represented by diamonds rather than duration bars. In most tools, you add a milestone by creating a new task and setting its duration to zero or selecting a dedicated "Add Milestone" option to display a specific, often gold-colored, marker on the timeline

How to Add Milestones

Dedicated Software (e.g., TeamGantt, GanttPRO): Click the "Add Milestone" button, or right-click an existing task and choose "Add milestone" or "Convert to milestone".

Microsoft Project: Select a task, choose "Task Information," and check "Mark task as milestone" or set the duration to 0 days.

Excel/Manual: Create a new row, set the start/end date to the same date, and change the format to a diamond shape.

Key Features

Dependencies: Link milestones to the completion of key tasks, so if a task delays, the milestone automatically updates.

Duration: While standard milestones have zero duration, you can add them to phases that take time, such as an approval process.

Visibility: Use distinct colors (e.g., gold) for milestones to make them immediately recognizable to stakeholders

Add a milestone

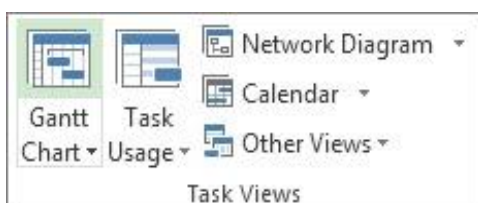
Applies To

Before your project gets rolling, you will probably want to mark its major goals with milestones.

Create a milestone with zero duration

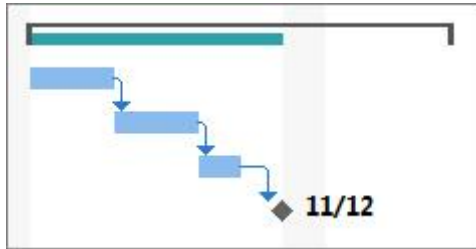
The quickest way to create a milestone is to add a task with no duration to your project plan.

1. Select **Task > Gantt Chart**.



2. Type the milestone name in the first empty row or pick a task you want to turn into a milestone.
3. Type **0** in the **Duration** field, and then press Enter.

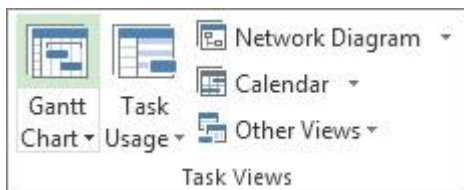
The milestone symbol is now part of your Gantt Chart.



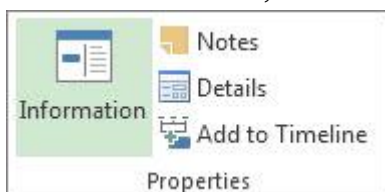
Add a milestone with a duration

Sometimes a milestone takes time. For example, the approval process at the end of a phase might take a week, so that milestone would need to take place over time like a normal task.

1. Select **Task > Gantt Chart**.

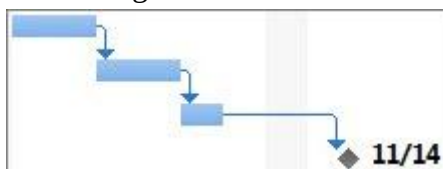


2. Type the milestone name in the first empty row or pick a task you want to turn into a milestone.
3. Select the milestone, and then select **Task**. In the **Properties** group, select **Task Information**.



4. Select the **Advanced** tab, and then type the milestone duration in the **Duration** box.
5. Check **Mark task as milestone**, and then select **OK**.

On the Gantt Chart, the milestone symbol appears on the last day of the task. It doesn't appear as a bar, even though it has duration.



Add an external milestone

Sometimes you might need a milestone to track a task that's outside the scope of your project.

If the milestone depends on a project that is beyond your control, such as software that is being developed by another company, create a milestone using the steps in the previous section. You'll have to keep an eye on the external task and update it manually.

Tracking Gantt Chart

A **Tracking Gantt Chart** is a specialized project management visualization used to compare a project's original schedule (the baseline) with its current status or actual performance. It allows project managers to immediately identify deviations, such as tasks that are falling behind or finishing early. A Tracking Gantt Chart is a specialized project management tool used to visualize the progress of a project by comparing the original planned dates (baseline) against actual execution dates. It allows managers to immediately identify schedule deviations, slippage, and whether tasks are ahead of or behind schedule.

Key Components of a Tracking Gantt

Tracking Gantt charts typically display two sets of task bars for each activity to facilitate direct comparison:

- **Baseline Bar:** Usually the lower bar, representing the original planned start and finish dates saved at the beginning of the project.

- **Actual/Scheduled Bar:** The upper bar, showing the current projected or actual start and finish dates.
- **Progress Shading:** Shading within the bars often indicates the percentage of completion for each task.
- **Variance:** The gap between the baseline and actual bars visually represents the schedule variance, helping to identify "slippage".

Benefits of Tracking Planned vs. Actual

- **Early Issue Detection:** Spot potential bottlenecks and delays early to take corrective action before they impact the entire project timeline.
- **Clear Stakeholder Communication:** Provides a presentation-ready visual to explain shifting schedules and impacts to clients or executives.
- **Improved Forecasting:** Analyzing historical variance data helps managers build more realistic timelines for future projects.
- **Accountability:** Clearly shows task owners and whether they are meeting their designated deadlines.

Key Techniques for Comparing Planned vs. Actual Dates:

- **Baseline Creation:** Before work begins, save the initial schedule as a "baseline" to act as a reference point. The Tracking Gantt view will display this as a lower, often grayed-out bar.
- **Visual Comparison:** The upper, colored bar represents actual progress (actual start/end dates). If the upper bar is longer or shifted right compared to the baseline, the task is delayed.
- **Progress Updates:** Update tasks with actual start and finish dates, or percentage completion. The chart dynamically updates to show if milestones are slipping.
- **Conditional Formatting (Excel):** Use formulas to highlight cells where the actual date extends beyond the planned date, often using color codes to indicate status (e.g., green for on track, red for delayed).
- **Identify Slipping Assignments:** Use specific filters in project software to show resources whose tasks are delayed beyond the baseline finish date.

How to Use a Tracking Gantt Chart

- **Set a Baseline:** Before work begins, save your planned schedule as a "baseline" in your project management software (like Microsoft Project or ProjectManager).
- **Update Progress Regularly:** As the project proceeds, enter actual start dates, actual finish dates, and percentage complete for each task.
- **Analyse Variances:** In tools like Microsoft Project, use the Tracking Gantt view and apply the Variance table to see calculated differences between planned and actual work.
- **Identify Critical Path Slippage:** Focus on tasks on the critical path, as any delay here will directly push back the final project completion date.

Critical Path Method for Project Management

Critical Path Method (CPM) is a project management technique used for planning, scheduling, and controlling projects. It helps determine the minimum time required to complete a project and identifies activities that directly impact the project completion date. CPM is especially useful for complex projects with multiple interdependent tasks.

- A critical task is an activity that cannot be delayed without delaying the overall project completion. Any delay in a critical task immediately affects the project deadline.
- The critical path is the longest sequence of dependent activities in a project network. It represents the shortest possible project duration. All activities on this path are called critical activities, and they have zero float.

What is Critical Path?

Critical Path deals with the path that includes all the critical project activities. Generally, it is defined as the longest sequence of activities in our project that has the shortest completion time.

1. It includes crucial as well as non-crucial tasks of the projects that allow us to prioritize the tasks as per their importance in the workflow.

2. The more the number of tasks, more the time is required to complete them.
3. Suppose, you have two ways to complete the project. One path contains all the tasks that form the major portion of the tasks and another one includes the minor and non-crucial tasks.
4. Thus, the path with major tasks will help us to deliver the project faster.
5. This is the reason why this longest sequence of the tasks has shortest delivery time.

Relationship Between Critical Task and Critical Path

A critical task (activity) and the critical path are closely related concepts in the Critical Path Method:

- A critical task is any activity with zero float, meaning it cannot be delayed without delaying the project.
- The critical path is the continuous sequence of critical tasks from the start of the project to the end.
- Every activity on the critical path is a critical task, but not every project activity is critical.
- Delaying any single critical task will delay the entire critical path, and therefore delay the project completion.

Benefits of Using Critical Path Method

- Provides a clear visual representation of the project schedule
- Identifies the most important (critical) tasks
- Helps anticipate and manage project risks
- Improves communication and coordination within the project team

Steps to Find the Critical Path

1. Identify all activities required to complete the project
2. Define the sequence and dependencies of activities
3. Estimate the duration of each activity
4. Draw the Activity-on-Arrow (AOA) network diagram
5. Perform a forward pass to calculate earliest times
6. Perform a backward pass to calculate latest times
7. Calculate float and identify the critical path

What is the Example of Critical Path Method in Project Management?

If you are looking for a step-by-step explanation of Critical Path, you do not need to look further. Our guide on the Critical Path covers how to implement this crucial concept in Project Management. By the end of the article, you will gain a clear understanding of the Critical Path with examples. So, you are on the right track to learn and implement the Critical Path Method for your Project. While managing a Project, professionals always focus on the practical implementation rather than the theoretical mastery of the topic. And, to find the Critical Path for Project too, we must gain a practical understanding so that we can identify the most crucial activities of the Project and manage the [project timeline](#). Thus, examples and scenario-based understanding will help you to implement the Critical Path Method in your Project.

In this insightful blog, we will quickly learn about Critical Path in [Project Management](#), not only using the definition but also some real-world examples. Let us start without any single point of delay.

Key Components of CPM

Before learning the entire process of using the CPM method, let us understand the components of the CPM which are described below.

1. **Work-Breakdown Structure:** It defines all the set of activities that are required for [project execution](#).
2. **Estimated Completion Time:** This means when the task is expected to be completed.
3. **Task Dependencies:** It defines what tasks are required to be executed before any other tasks.
4. **Earliest Start Time (ES) and Finish Time (EF):** When the project starts, how early we can start, and how early we can finish is determined using these parameters.
5. **Latest Start Time (LS) and Finish Time (LF):** When the project is in progress, the Latest start time is the latest point when we can start the task without delaying it and the Latest Finish Time is the time when we can finish it without delaying it.

6. **Float:** This is the duration for which a delay in task execution is allowed in a project.

Applying CPM in Practice: A Step-by-Step Guide

Now, let us understand the process of using the Critical Path in the Project. This concept is important for two reasons. First is that it will help you to understand Critical Path more deeply, and second is that you can analyze these steps to understand critical path examples and implement them for your use-case. The path to finding the critical path is called the Critical Path Method (CPM). A typical process to find the Critical Path in the Project is discussed as shown below.

Step 1: Find what tasks to do for the project

Your Project may include different tasks related to end-user experience, designing the architecture, essential functional features, testing of the tasks, etc. Project Tasks are not just limited to this. There can be more tasks. These tasks, processes, and activities have to be identified so that we can get a high-level idea of the project.

Step 2: Find the sequence of the tasks

After you establish a high-level understanding of the tasks, you can now identify their flow. It means that you have to list what task should be performed and when. Here, you also have to identify the dependencies among the different tasks. Thus, you can easily know which task has to be performed first so that an accurate workflow can be set up.

Step 3: Know how long the tasks will take to complete

Now, you have to find the estimated completion Time for each task which is the total time from its start to the completion. It is important to find the estimation time with accuracy so that the precise flow of the [project activities](#) can be determined.

Step 4: Draw a Network Diagram to find multiple ways

The Project Network Diagram is the visual representation of the various possible paths from the beginning of the project to the completion of the project. It includes a Node that tells us about the various parameters of the Task such as Earliest Start Time, Earliest Finish Time, Latest Start Time, Latest Finish Time, and Float.

Step 5: Find the optimal way which is the Critical Path

Now, analyze different paths and then choose the optimal path. You have to find the path that contains the longest sequence of the project tasks and the minimum completion time for the project. The Critical Path that we selected contains all the crucial interdependent tasks in our project.

Step 6: Find which tasks can be delayed and how much

By finding the Critical Path, we are aware of the crucial tasks. Now, you have to analyze the non-crucial tasks which can be delayed. You also have to calculate their delay time without impacting the quality of the project.

The above steps have to be performed to find the critical path. Now, let us see practical examples of Critical Path in Project Management.

Examples of CPM

Suppose that we have to manage a project of E-commerce Software Development. Then, to find the critical path of the project. Before delving deep into the critical path examples, let us understand the following terminologies.

1. Define the Work Breakdown Structure (WBS)

As we already know it includes all the activities in our project. Thus, we have to create a WBS that covers various tasks that need to be performed for the E-commerce Software Development. The tasks for this example will be

1. User Survey
2. Stakeholder Opinions
3. Market Research
4. Requirements Gathering
5. UI/UX Design
6. Database Design
7. Front-end Development
8. Back-end Development

9. Integration and Testing

10. Deployment

2. Calculate Parameters for Task Duration

This operation is generally performed by the project professionals to assign the duration of completion to each task. Here, various calculated parameters are shown in the below table including the formulas to create them:

Calculated Parameters with Formulas

	Task	ES (weeks)	EF (weeks)	LS (weeks)	LF (weeks)	Duration (weeks)	Duration (weeks) Total Float (weeks) (LS-ES)
1	User Survey	0	2	0	2	2	0
2	Stakeholder Opinions	EF of Task 1	2	EF + Duration	3	1	0
3	Market Research	EF of Task 2	3	EF + Duration	6	3	0
4	Requirement s Gathering	EF of Task 3	6	EF + Duration	8	2	2
5	UI/UX Design	EF of Task 4	8	LF- Duration	12	4	0
6	Database Design	EF of Task 5	12	LF- Duration	15	3	0
7	Front-end Development	EF of Task 6	15	LF- Duration	21	6	0
8	Back-end Development	EF of Task 7	15	LF- Duration	20	5	1
9	Integration and Testing	EF of Task 8	21	LF- Duration	25	4	0
10	Deployment	EF of Task 9	25	LF- Duration	27	2	2

3. Find the Critical Path

As seen, the zero float time of tasks denotes that they are critical for the project and cannot be delayed. Therefore, the critical path for our project will be 1-2-3-5-6-7-9. And, the float which we have calculated allows us to manage the tasks which can be delayed and executed later.

Benefits of CPM

1. **It gives us a clear project visualization:** Critical Path highlights the important tasks and their dependencies. Thus, we can easily visualize the project execution to ensure successful delivery.
2. **Resources are not wasted:** Since we are only allocating resources for crucial tasks, the resource allocation becomes efficient. As a result, project costs can be easily reduced.
3. **We can also define the project baseline:** The project baseline includes various indicators that indicate the accuracy of the project. CPM helps to highlight it so that we can take our project in the right direction.

Limitations of CPM

1. **We have to assume fixed activities:** CPM considers that activities are fixed which is not applicable in all the cases. Due to this rigid nature, we may face problems while adapting to the dynamic environment.
2. **Non-critical tasks are tough to manage:** CPM focuses more on critical tasks and less on non-critical tasks. This may lead to delays in the project activities. Hence, the overall timeline is disturbed.
3. **It is challenging to handle the uncertain changes:** Once the project activities are defined, there is less room for uncertain or sudden changes. Thus, CPM is not suitable for an environment where changes or updates are too frequent.

How to Identify the Critical Path?

One can pinpoint the critical path by evaluating the lengths of critical and non-critical tasks. Below are examples of each stage.

1. Listing Your Activities

Create a work breakdown structure (WBS) to compile a comprehensive list of all project activities and tasks needed for deliverables. The subsequent Critical Path Method (CPM) relies on this activity list.

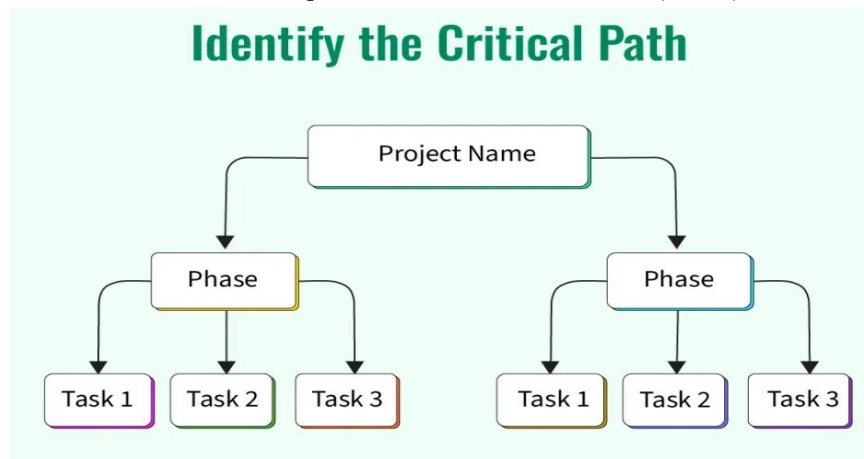


Illustration:

Consider the scenario where the education department is crafting a new interactive blog article. Sample tasks in the WBS might include:

Task ID	Task	Duration (in days)
A	Draw up a plan	1
B	Create a draft	3

Task ID	Task	Duration (in days)
C	Edit the draft before submitting it.	2
D	Graphics for design posts	5
E	Visuals can be animated	3
G	Put up articles	2

On Starting up how we can do it let's try to identify task dependencies as possible once you have to give your general understanding of everything that needs to be done.

2. Identifying Dependencies

Select interdependent jobs based on the work breakdown structure. Identify tasks that can run concurrently.

Example:

Dependencies for the tasks mentioned:

1. A is needed for Task B.
2. B is needed for Task C.
3. Tasks C and D can run simultaneously.
4. D is needed for Task E.
5. C, D, and E all impact Task F.

The list of dependent tasks forms the critical path.

3. Creating a Network Diagram

Build a network diagram, a flowchart displaying event sequences, using the work breakdown structure. Each task is represented by a box with arrows indicating dependencies. Before finalizing the project schedule, add time-bound components to the network diagram.

A well-drawn network diagram is the project's architectural blueprint. It's the visual guide for tasks, connections, and the critical path. Proper Think about it when we want it as a map the team can follow from beginning to finish.

4. Estimating Task Duration

To compute the critical path, determine the duration of each activity. The critical path is the longest series of critical tasks.

Estimation Methods:

1. Based on experience and understanding.
2. Using data from previous projects.
3. Utilizing industry norms as estimates.
4. Consider the forward pass and backward pass method.

5. Calculating the Critical Path

While it's possible to manually determine the critical path, using a critical path algorithm is faster.

Manual Steps:

Step 1: Note the start and end times for each activity.

1. The duration of the first activity is its start time (0) and finish time.
2. The time that is going to end becomes the start for the next, and the finishing time is the sum of the two also.
3. Repeat for each activity.

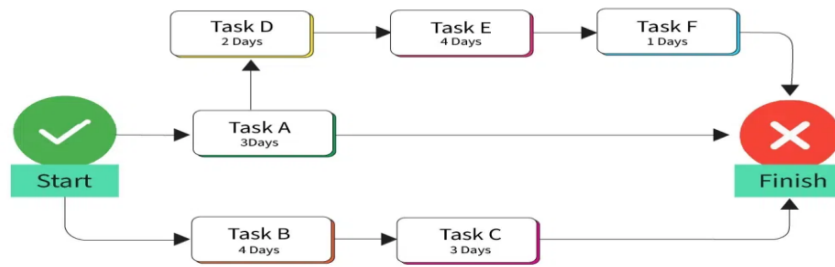
Step 2: Look at the final activity's end time to calculate the sequence's length.

Step 3: The critical path is the set of tasks taking the longest to complete.

Example:

Critical path diagram

Critical path diagram



Construct the actual project schedule around the identified critical path.

6. Calculating the Float

Task slack, or float, indicates a task's flexibility. It represents the time that can be added without impacting other tasks or the project deadline. Determine the float to gauge project flexibility and cover potential risks. Manual or algorithmic methods can be used to calculate total and free float.

Total Float: The duration that can pass since the early start date without postponing project completion or violating a schedule constraint.

Total Float = LS-ES or LF-EF

Free Float: The maximum time an activity can be delayed without affecting the next. It applies when two or more activities converge on a single successor in a network diagram.

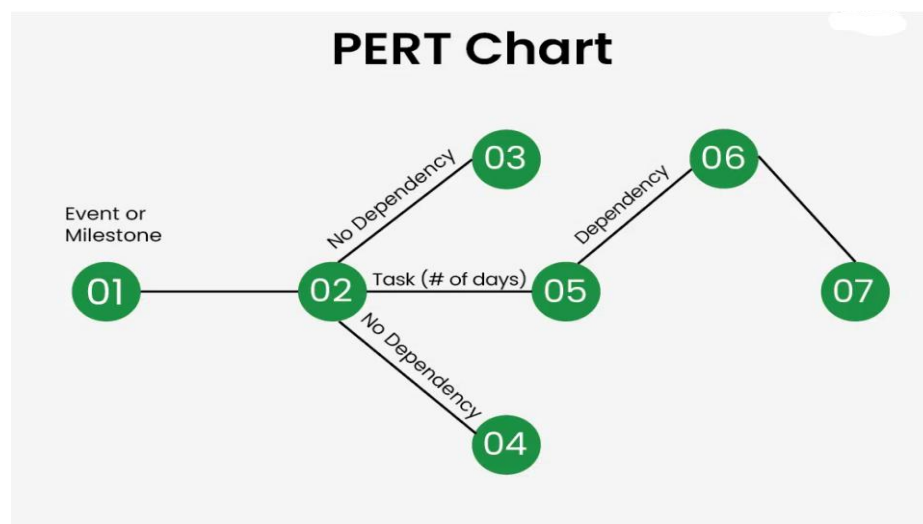
Free Float = ES (next task) - EF (current task)

Project Managers Benefit from Understanding Float:

1. Ensures projects finish on schedule.
2. Helps prioritize tasks.
3. A valuable tool for addressing risks or unforeseen problems effectively.
4. Understanding these concepts contributes to successful project management.

Project Evaluation and Review Technique (PERT)

Project Evaluation and Review Technique (PERT) is a procedure through which activities of a project are represented in its appropriate sequence and timing. It is a scheduling technique used to schedule, organize and integrate tasks within a project. PERT is basically a mechanism for management planning and control which provides blueprint for a particular project. All of the primary elements or events of a project have been finally identified by the PERT. In this technique, a PERT Chart is made which represent a schedule for all the specified tasks in the project. The reporting levels of the tasks or events in the PERT Charts is some what same as defined in the [work breakdown structure \(WBS\)](#).



What is PERT Chart?

A PERT chart is a project management tool used to plan and schedule tasks, illustrating the sequence and timing of project activities. The PERT chart is used to schedule, organize and co-ordinate tasks within the project. the objective of PERT chart is to determine the critical path, which comprises critical activities that should be completed on schedule. This chart is prepared with the help of information generated in [project planning activities](#) such as estimation of effort, selection of suitable process model for [software development](#) and decomposition of tasks into subtasks.

What does a PERT Chart Contain?

Here are the main components of a PERT chart:

- **Nodes:** it represents the task or milestones. every node represents the task name and may also show duration of the task.
- **Arrows:** it indicates the direction or sequence of task and also dependencies between them. suppose an array from A to B, then task A must be completed before task B.
- **Time Estimation:** It estimates the time duration to complete the task.
- **Critical Path:** The critical path is the largest path in [project management](#) that always results in the shortest time to complete the project.
- **Milestones:** It is Key point in the project timeline that represent significant events or deadlines.

How a PERT Chart Works?

A **PERT** chart used to plan and visualize tasks in a project. It breaks down the project into individual tasks and shows the sequence in which they must be completed. Each task is represented by a node, and arrows indicate the dependencies between tasks. By analyzing the chart, teams can identify the critical path, which helps determine the shortest time to complete the project and allocate resources effectively.

How to Create a PERT chart?

To create a PERT chart, we can follow the below steps:

Step 1: Identify Project Tasks

List all the tasks required to complete the project. Identifying tasks means listing all the steps needed to complete a project. Start by figuring out what the main goal is, then break it down into smaller actions. Ask team members for ideas and use clear action words for each task.

Step 2: Define task dependencies

Defining task dependencies means figuring out which tasks must be completed before others can start. Look at your task list and identify connections by asking if one task relies on another. This helps you establish the order of work, ensuring everything is done in the right sequence.

Step 3: Estimate Timeline

Estimating the timeline involves figuring out how long each task will take to complete. For each task, think about the quickest time it could be done (optimistic), the longest it might take (pessimistic), and the most likely time it will actually take. This gives you a better idea of the overall project duration and helps with planning.

Step 4: Calculate Critical Path

Calculate [Critical Path](#) means finding the longest sequence of tasks that determines the shortest time to complete the project. Identify which tasks can't be delayed without affecting the overall [project deadline](#), helping you focus on what's most important.

Step 5: Manage task progress

It involves tracking how each task is going. Regularly check if tasks are on schedule, address any delays, and adjust plans if needed. This ensures the project stays on track and any issues are resolved quickly.

What is PERT vs CPM?

Here are the following differences between PERT and CPM:

Aspect	PERT	CPM
Abbreviation	PERT stands for Project Evaluation and Review Technique.	CPM stands for Critical Path Method
Definition	PERT is a technique of project management which is used to manage uncertain (i.e., time is not known) activities of any project.	CPM is a technique of project management which is used to manage only certain (i.e., time is known) activities of any project.
Orientation	It is an event-oriented technique which means that network is constructed on the basis of event.	It is an activity-oriented technique which means that network is constructed on the basis of activities.
Model Type	It is a probability model.	It is a deterministic model.
Focus	It majorly focuses on time as meeting time target or estimation of percent completion is more important.	It majorly focuses on Time-cost trade off as minimizing cost is more important.
Precision	It is appropriate for high precision time estimation.	It is appropriate for reasonable time estimation.
Nature of Job	It has Non-repetitive nature of job.	It has repetitive nature of job.
Crashing	There is no chance of crashing as there is no certainty of time.	There may be crashing because of certain time bound.
Dummy Activities	It doesn't use any dummy activities.	It uses dummy activities for representing sequence of activities.
Sustainability	It is suitable for projects which required research and development.	It is suitable for construction projects.

Characteristics of PERT Chart

The main characteristics of PERT are as following:

- It serves as a base for obtaining the important facts for implementing the decision-making.
- It forms the basis for all the planning activities.
- PERT helps management in deciding the best possible resource utilization method.
- PERT take advantage by using time network analysis technique.
- PERT presents the structure for reporting information.

- It helps the management in identifying the essential elements for the completion of the project within time.
- It specifies the activities that form the critical path.
- It describes the probability of completion of project before the specified date.
- It describes the dependencies of one or more tasks on each other.
- It represents the project in graphical plan form.

Advantages of PERT Chart

It has the following advantages:

- Estimation of completion time of project is given by the PERT.
- It supports the identification of the activities with slack time.
- The start and dates of the activities of a specific project is determined.
- It helps project manager in identifying the critical path activities.
- PERT makes well organized diagram for the representation of large amount of data.

Disadvantages of PERT Chart

It has the following disadvantages:

- The complexity of PERT is more which leads to the problem in implementation.
- The estimation of activity time are subjective in PERT which is a major disadvantage.
- Maintenance of PERT is also expensive and complex.
- The actual distribution of may be different from the PERT beta distribution which causes wrong assumptions.
- It under estimates the expected project completion time as there is chances that other paths can become the critical path if their related activities are deferred.

Planning Cost Management:

What is a Cost Management Plan?

A cost management plan is a document that helps you map and control a budget. It enables project managers to estimate their costs, allocate resources to the right areas, and control overall spending.

Cost management plans keep all project costs in one place, including direct and indirect costs. A project manager will track these costs to ensure there are no budget overruns.

A cost management plan example could be the budget for a home improvement project. Direct costs would include hired labour and building materials. Indirect costs would include equipment rental fees, insurance, and general maintenance.

What is included in a cost management plan?

For effective cost management in project management, a project manager needs to ensure they have all the required information to create a detailed plan.

A cost management plan can include many different elements, such as:

- **Measurement units:** These will differ depending on the region. If you are in the US, you'll use inches; if you are in Germany, you'll use meters.
- **Precision levels:** This is where you determine the number of decimal places in your calculations. You can also define the amount of flexibility in your deadlines and figures.
- **Control thresholds:** Set a threshold for taking action if you are behind on schedule. If you hit the -\$5,000 mark, for example, this can be the point where you correct course to get back on track.
- **Performance measurement rules:** Consider the techniques for measuring your performance. Will you use earned value management or the percentage of completion method?
- **Reporting formats:** Determine how you will report your project progress and how often.

Why is a cost management plan important?

A cost management plan is important because it creates a detailed outline of your project budget. It helps project managers to stay organized and on track to hit their project deliverables.

If you don't estimate costs accurately, allocate sufficient resources where they are needed, and regularly monitor spending, you run the risk of budget overrun. This can lead to project failure. Therefore, cost planning in project management is an essential exercise in ensuring project success.

A project cost management plan can also be highly beneficial for cutting unnecessary costs. One of the biggest challenges project managers face is an overstretched budget with no clear indication of excess spending. According to Harvard Business Review, “to cut costs effectively, companies must connect costs to their strategy.” A cost management plan will help project managers to assess cost priorities and save money.

How to make a cost management plan

To make a project cost management plan, you need to follow four simple steps:

1. **Break work into tasks:** Build a work breakdown structure with folders and subfolders, which you can break down further into tasks and subtasks. A useful tool such as Wrike Resource will help you pick the right person for each task.
2. **Estimate your costs:** Create a cost estimation forecast for your project. Include your direct and indirect costs, cut unnecessary spending, and budget for unforeseen events. Ensure you use flexible software as your forecast is subject to change.
3. **Create your budget:** Once your cost estimation is complete, you can go into more detail with your actual budget. Use milestones in Gantt charts to separate your budget into stages for easier cost management.
4. **Monitor performance:** A cost control plan is vital in tracking performance. Match team progress to your expenses to date and ensure you are on schedule.

Types of Cost Estimates

Not all cost estimates are created equal. In fact, project managers typically develop different types of estimates at different stages of a project. Early in the project (when many details are still unknown), you may provide a rough figure just to gauge feasibility. Later, once the scope is clearer, you'll prepare a detailed breakdown. Here are the two main types of cost estimates to consider in project management

1. Ballpark (Rough) estimate

A ballpark estimate is a high-level approximation of cost, often given as a range, used in the very early stages of a project. Ballpark estimates are typically based on limited information.

For example, a client asks, “*About how much would it cost to implement a new website?*” and you might respond with a broad range (say, “somewhere between \$80,000 and \$140,000”) based on experience with similar projects. The purpose of a ballpark estimate is to give stakeholders a general idea of the scale of the budget required before investing time in detailed planning.

Ballpark estimates are useful when the project scope is not yet fully defined. They help set initial expectations and facilitate early budget discussions. However, they are *not very precise*. It's understood that a ballpark figure may change significantly once more facts are known.

2. Definitive (Detailed) estimate

Sometimes called bottom-up estimating. With this method, detailed estimates are made at relatively low levels in the work breakdown structure (WBS), typically at work-package or task level. This approach is closely related to scheduling, planning and resource allocation and is both time-consuming and costly. It requires a good knowledge of the activity and there also needs to be a reasonable level of definition for the exercise to be meaningful. definitive estimate, sometimes referred to as a detailed estimate or Statement of Work (SOW) estimate, is a much more precise calculation of project costs. This type of estimate is prepared once the project scope and requirements are clearly defined and you have detailed information on deliverables, tasks, and resource needs.

The definitive estimate serves as a financial blueprint for executing the project. It should include all labour costs, material costs, external vendor fees, travel or logistics expenses, and even contingency reserves for risk. Because it is based on well-defined scope and task estimates, a definitive estimate is much more reliable than a rough estimate (the expected variance is much smaller, perhaps within $\pm 5\text{--}10\%$ if done correctly).

Cost Estimation Tools and Techniques:

Estimating the project cost can be a scary exercise for many managers. It is often seen that many project managers across the industries burn the midnight oil to prepare a perfect estimation. Despite all the efforts, when the project cost exceeds the estimated budget, they face a lot of flak from all corners. Though a perfect and accurate estimation of project cost may sound like climbing onto the peak of the

Himalayas, but a near-perfect, appropriate estimation can be done with the help of few proper tools and techniques.

There are a lot of uncertainties associated with the project in terms of schedule, scope, and resources. You do not have a clear picture at the beginning; the story unfolds in the later stages. So, it is important to access the future activities of your project at the initial phases considering the available information. Let us discuss the top 10 tools and techniques which are popular and are used for estimation –

Expert Judgement

While estimating the project cost, the first step is to take the comments from the experts. The experts are the people who have prior knowledge on similar kind of projects. So they can suggest valuable insight based on their experience. You can also take their advice on various tools and techniques that can be used to estimate similar kind of project.

Analogous Estimation

Normally, at the early stages of your project, you do not have much detail, so taking into account of similar projects previously completed by your organization, the cost of the project can be estimated. Analogous estimation technique uses the parameters such as scope, budget, duration, size, weight and complexity of previous projects having similar nature of works. It measures the current project on that basis and does the estimation.

For example, if the budget of a particular activity in the previous project has X amount, and by measuring the same activity in your current project looks identical, then the same X amount can be applied to that.

Parametric Estimation

This technique uses an algorithm to calculate the cost of the activity considering the historical data and other project variables. A statistical relationship needs to be evaluated between the historical data and other variables. This technique can be used for the complete project or for some of the activities in conjunction with other estimation techniques. This is one of the most accurate techniques to estimate the cost of the project.

For example, in an industrial project, one of the activities is to make 10 valves in the first phase. So as part of historical data gathering, you got the information that construction of each valve requires \$150. Based on that information you can calculate as:

Total cost of making 10 valves = cost per valves * no. of valves

Total cost = $150 \times 10 = 1500$

Bottom-Up Estimation

Bottom-up estimation technique starts with the estimation from the lower level i.e. the work package level created as per WBS and then rolled up to higher-level. The accuracy of this estimation technique is high as you are doing the estimation from granular level

Three-Point Estimation

To deal with uncertainties and risk, you need to take the help of three-point estimation which is also referred as PERT- Program Evaluation and Review Technique.

The Program Evaluation and Review Technique (PERT) use three types of estimations:

M – Most Likely: The realistic or ideal situation, all the required resources will be assigned and can achieve the expected productivity

O – Optimistic: Estimation based on best case scenario

P – Pessimistic: Estimation based on worst case scenario

Based on the above assumptions, the expected duration can be calculated using two basic formulas.

Triangular Distribution – $(O+M+P)/3$

Beta distribution – $(O+4M+P)/6$

Typical Problems with IT Cost Estimates

Cost estimation in IT projects is a difficult task because software projects are complex and uncertain. Many problems arise during estimation, which can lead to project failure, delays, or budget overruns.

1. Inaccurate or Incomplete Requirements

- At the beginning of a project, requirements are often not fully clear.

- Clients may change their needs during development.
- This leads to incorrect cost estimation.

Example: A client later asks for additional features not included initially → cost increases.

2. Optimism Bias (Overconfidence)

- Developers assume that everything will go smoothly.
- They ignore possible delays, bugs, or challenges.
- This results in underestimating cost and time.

Example: Estimating 2 weeks for a task that actually takes 1 month.

3. Lack of Experience

- Inexperienced team members may not understand project complexity.
- They may forget to include testing, maintenance, or integration costs.

Result: Unrealistic and inaccurate estimates.

4. Poor Project Scope Definition (Scope Creep)

- If the project scope is not clearly defined, new features keep getting added.
- This is called **scope creep**.

Example: Adding extra modules like payment gateway, reports, etc. after estimation.

5. Ignoring Risks

- Many estimates do not consider risks such as:
 - Technical issues
 - Staff turnover
 - Hardware/software failure
- These risks increase project cost later.

6. Lack of Historical Data

- Accurate estimation needs past project data.
- Without it, estimates are based on guesses.

Example: No previous data → difficult to estimate effort for a similar project.

7. Time Pressure for Estimation

- Managers often demand quick estimates.
- Due to lack of time, detailed analysis is not done.

Result: Rough and inaccurate cost estimation.

8. Technology Changes

- New or unfamiliar technologies create uncertainty.
- Teams may need extra time for learning and implementation.

Example: Using a new framework increases development time.

9. Resource Availability Issues

- Assumptions about availability of skilled people may be wrong.
- Delays occur if required resources are not available.

Example: Key developer leaves project → cost increases.

10. Poor Communication

- Miscommunication between client, manager, and team leads to misunderstanding.
- Requirements and expectations are not clearly shared.

Result: Wrong estimation and rework.

11. Hidden Costs Ignored

- Some costs are often not included:
 - Training
 - Maintenance
 - Documentation
 - Testing
- This leads to underestimation.